

Steepest descent - Method

$$f(x) = 4x^3 + 6x^2 - 9x + 1$$

find the minima

$$f(x) = 4x^3 + 6x^2 - 9x + b$$

$$a \rightarrow 4 \text{ to } 10$$

$$b \rightarrow 1 \text{ to } 10$$

learning rate $\alpha = 0.01$

Take initial guess $x = 0$

Iteration 1

$$f'(x) = 12x^2 + 12x - 9$$

$$= \boxed{-a}$$

$$x = x - \alpha f'(x) = \boxed{} \checkmark$$

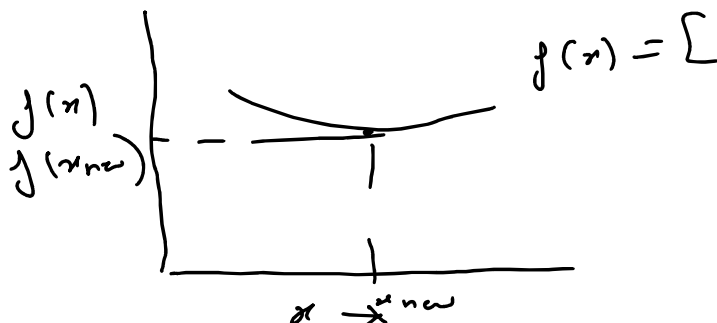
Iteration 2

$$f'(x) = 12x^2 + 12 - 9 = \boxed{}$$

$$x = x - \alpha f'(x) = \boxed{} \checkmark$$

	x	$f'(x)$	x_{new}
1	0	$\checkmark$	$\checkmark$
2	x_{new}	$\checkmark$	$\checkmark$

$$10 \quad \boxed{x_{new}} \quad x = [0, 0.1, 0.2, \dots, 2]$$



$$\text{Minima } f(x) = \boxed{} \checkmark$$

